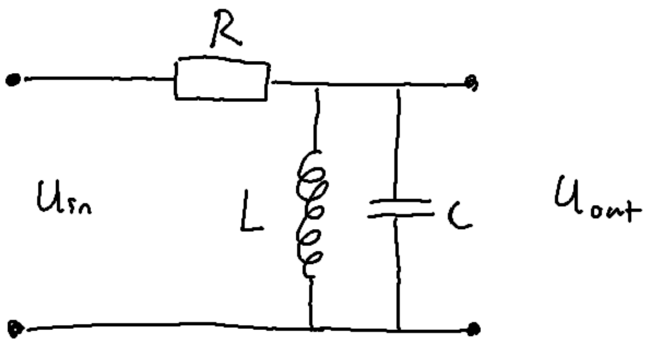
electronics 2

1

2

$L = 10mH, \qquad C = 100nF, \qquad R = 1k\Omega, \qquad U_{\text{in}} = 1V \quad (2.8V_{\text{pp}})$



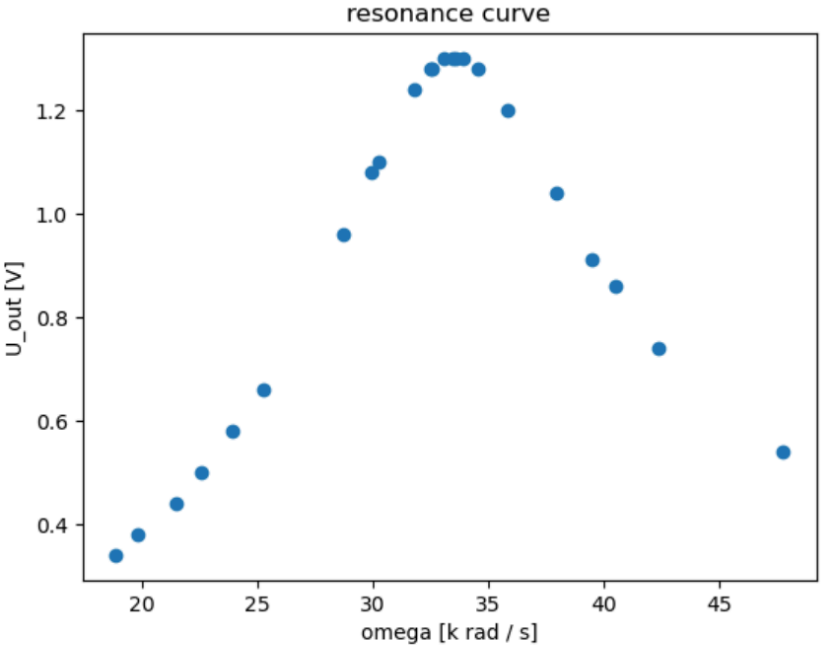
2.1

for the phase shift between U_in and U_out we used math function

we also calculated:

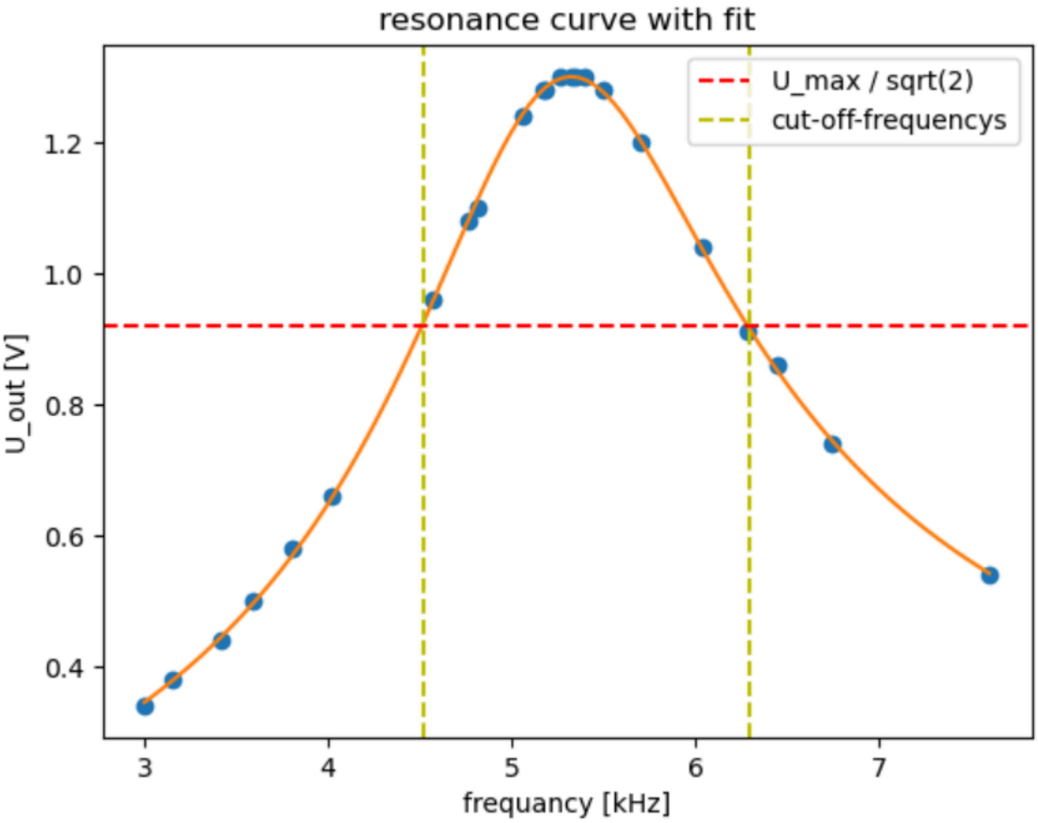
$$\frac{U_{\text{out}}}{U_{\text{in}}} = \left(R^2\left(\omega C - \frac{1}{\omega L}\right)^2 + 1\right)^{-\frac{1}{2}} \Rightarrow \omega_0 = \frac{1}{\sqrt{CL}} \tag{2.1}$$

2.2



$$\Rightarrow \omega_0 \approx 34000\frac{\text{rad}}{s} \tag{2.2}$$

2.3



$$Q_{\text{theory}} = R\sqrt{\frac{C}{L}}\ \text{(manual)} \tag{2.3}$$

since Equation (2.1) we used the following fit:

$$f(\omega, a, b, c) = \left(c + \left(\omega b - \frac{1}{\omega a}\right)^2\right)^{-\frac{1}{2}} \Rightarrow \omega_0 = \frac{1}{\sqrt{ab}} \tag{2.4}$$

and got the fit parameters: $a \approx 0.081, b \approx 0.43, c \approx 0.59$ where the inputvariable ω we used for kHz. the equation:

$$\frac{f(\omega_0, a, b, c)}{\sqrt{2}} = \frac{U_{\text{max}}}{\sqrt{2}} == f(\omega, a, b, c) \tag{2.5}$$

gives the quadratic:

$$\omega^2 b - \omega\sqrt{\frac{2}{U_{\text{max}}^2} - c} - \frac{1}{a} = 0 \tag{2.6}$$

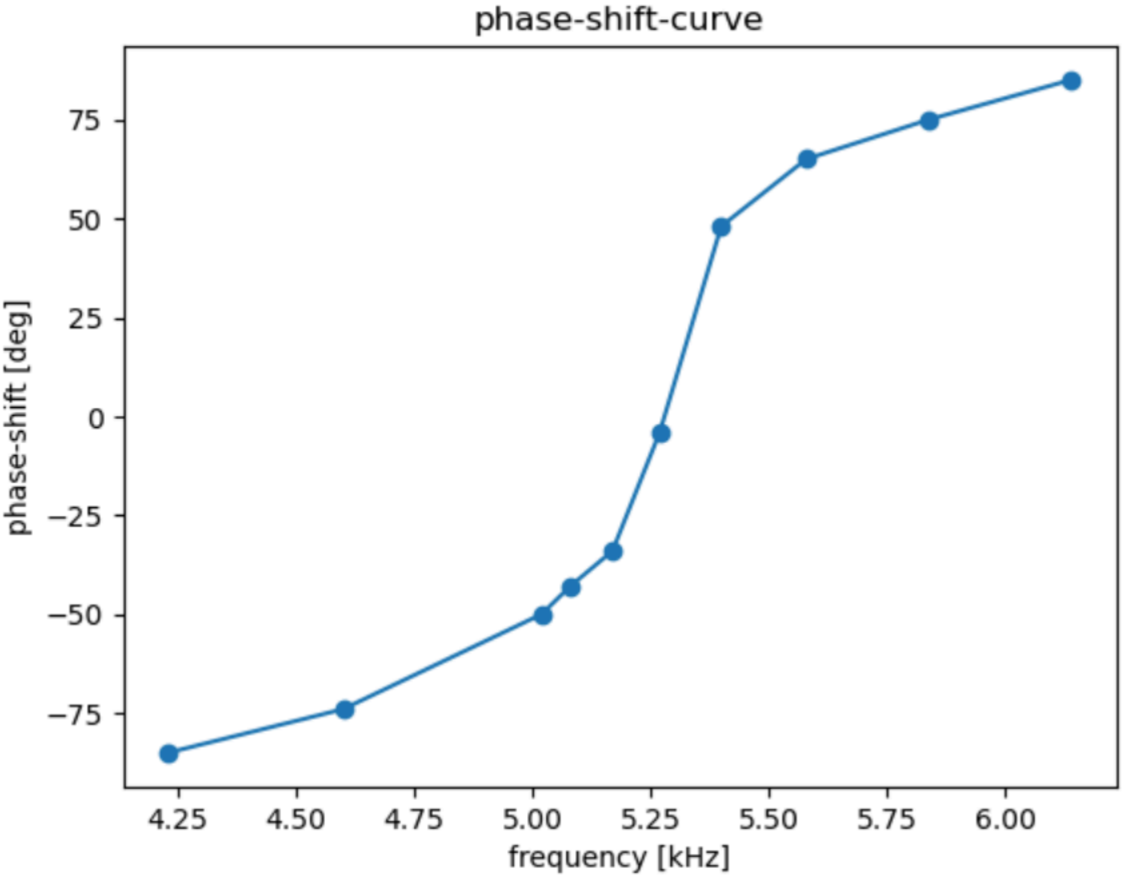
this yields:

$$\omega_{\text{lower}} \approx 4.51 * 2 * \pi \quad k\frac{\text{rad}}{s}, \qquad \omega_{\text{upper}} \approx 6.23 * 2 * \pi \quad k\frac{\text{rad}}{s} \tag{2.7}$$

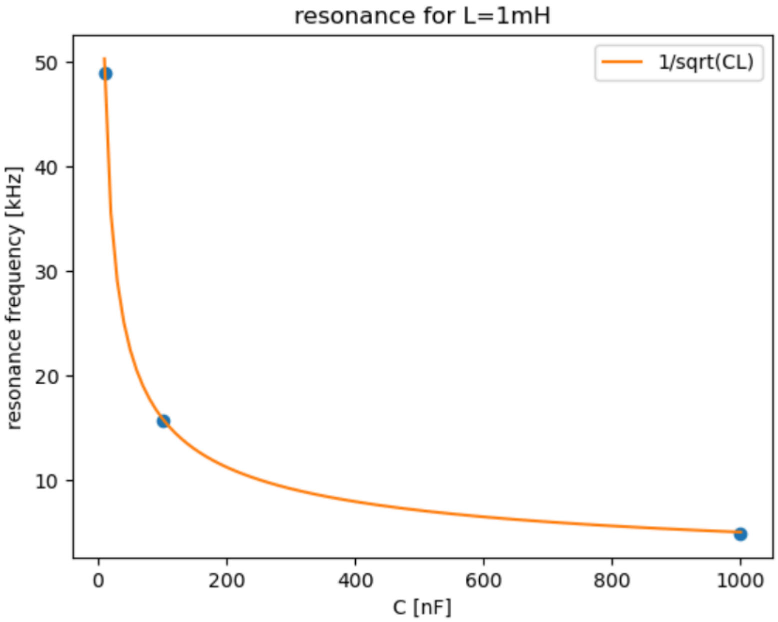
$$Q_{\text{fit}} = \frac{\omega_0}{\Delta\omega} = \frac{1}{\sqrt{ab}}\frac{1}{(\omega_{\text{upper}} - \omega_{\text{lower}})} \approx 3.00 \tag{2.8}$$

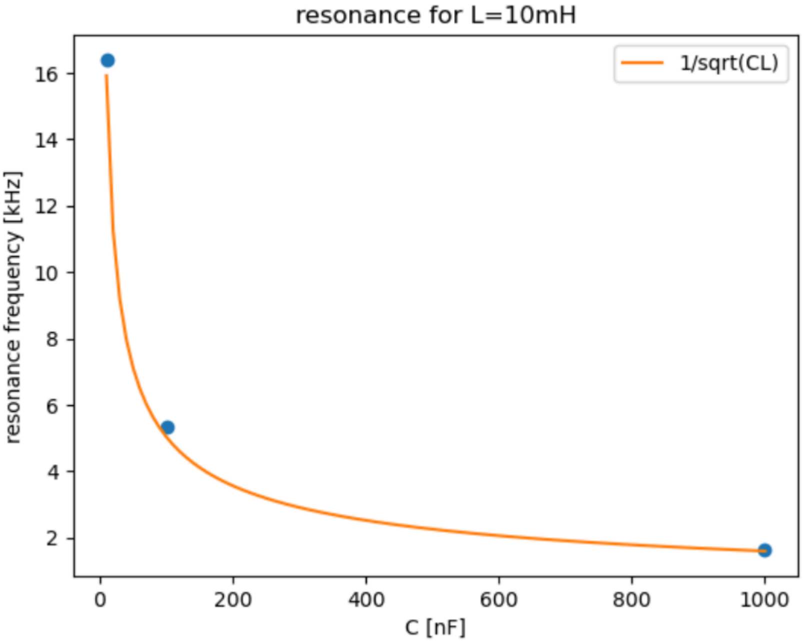
2.4

To measure the phase of the current I and U_{out} we measured the voltage over R U_R, since it is in phase with th current. To measure U_R we did $U_{\text{in}} - U_{\text{out}}$, so ch-1 - ch-2. We did this with the math function on the oscilloscope. To get a reading for the phase we did phase beween math and ch-2, which is U_{out}.



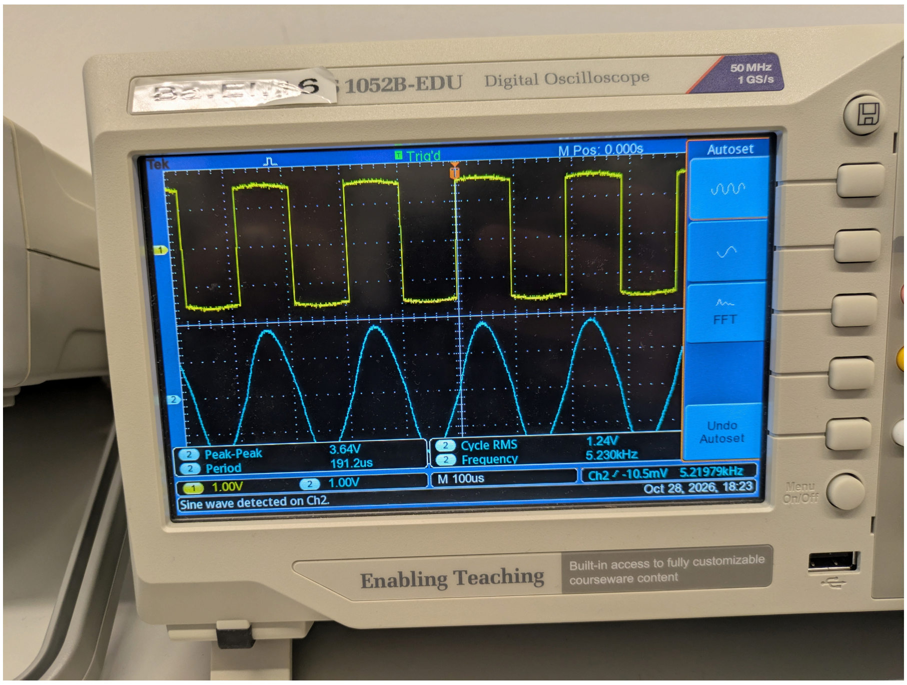
2.5





from Equation (2.1) we get $\omega_0 = \frac{1}{\sqrt{LC}}$

2.6


1052B-EDU Digital Oscilloscope
50 MHz 1 GS/s
Trig'd
M Pos: 0.000s
Autoset
FFT
Undo Autoset
Peak-Peak 3.64V
Period 191.2us
Cycle RMS 1.24V
Frequency 5.230kHz
1.00V
1.00V
M 100us
Ch2 -10.5mV 5.21979kHz
Oct 28, 2026, 18:23
Sine wave detected on Ch2.
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Built-in access to fully customizable courseware content

Figure 1: square wave at ≈ 5 kHz

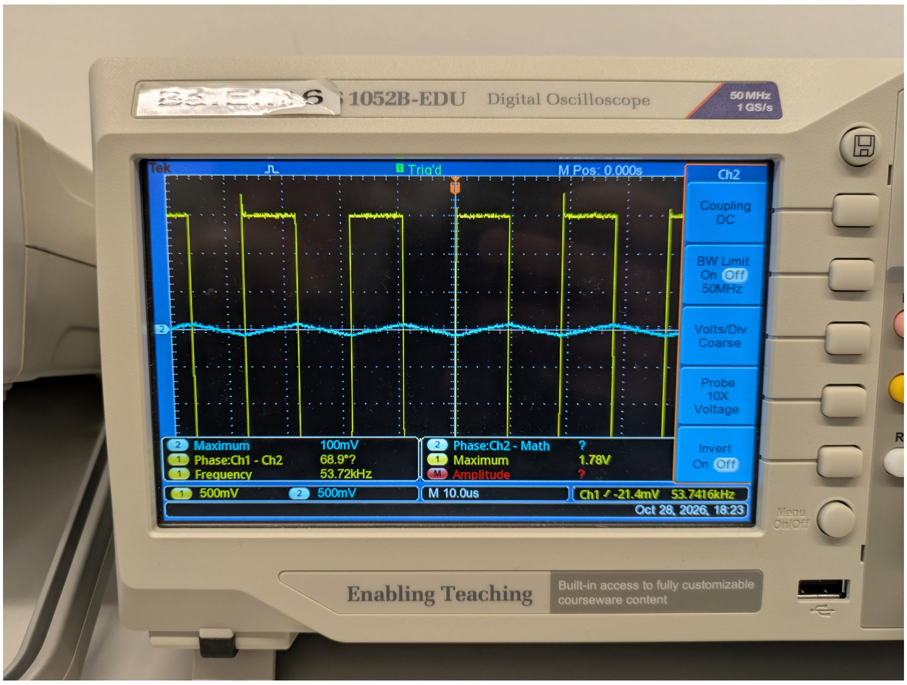
1052B-EDU Digital Oscilloscope
50 MHz 1 GS/s
Trig'd
M Pos: 0.000s
Ch2
Coupling DC
BW Limit On Off 50MHz
Volts/Div Coarse
Probe 10X Voltage
Invert On Off
Maximum 100mV
Phase:Ch1 - Ch2 68.9°?
Frequency 53.72kHz
Phase:Ch2 - Math ?
Maximum 1.78V
Amplitude ?
500mV
500mV
M 10.0us
Ch1 -21.4mV 53.7416kHz
Oct 28, 2026, 18:23
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Figure 2: square wave at ≈ 50 kHz

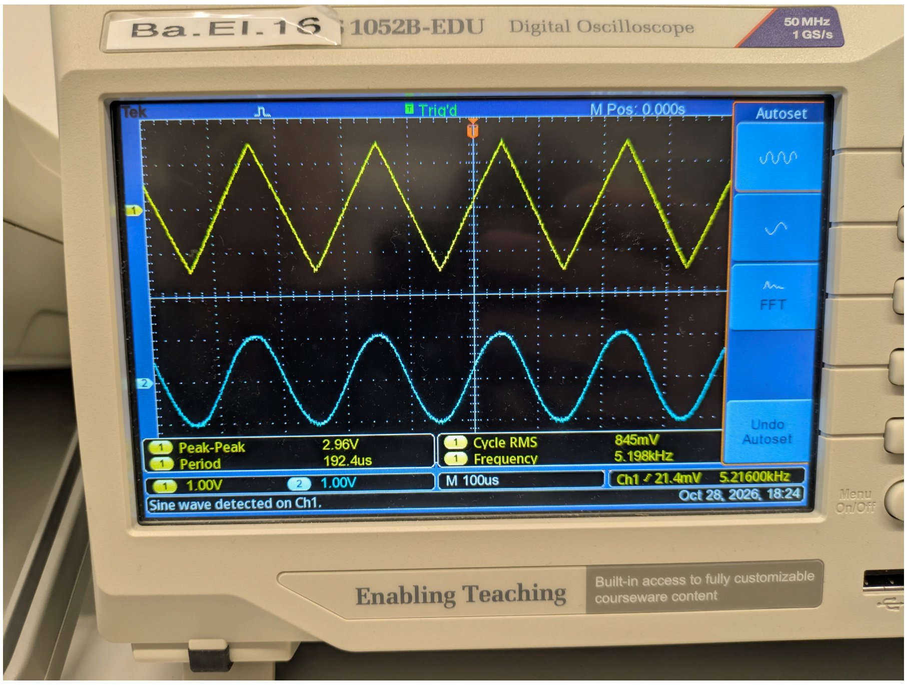
Ba.El.16
1052B-EDU Digital Oscilloscope
50 MHz 1 GS/s
Tek
Trig'd
M Pos: 0.000s
Autoset
FFT
Undo Autoset
Peak-Peak 2.96V
Period 192.4us
Cycle RMS 845mV
Frequency 5.198kHz
1.00V
1.00V
M 100us
Ch1 ⸝ 21.4mV 5.21600kHz
Oct 28, 2026, 18:24
Sine wave detected on Ch1.
Menu On/Off
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Figure 3: triangle wave at ≈ 5 kHz

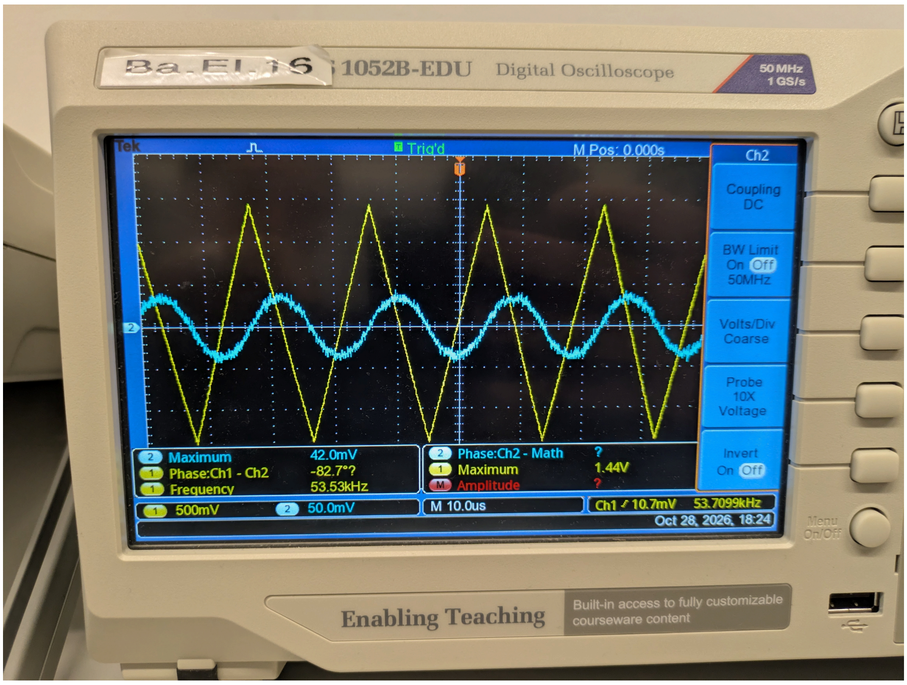
Ba.El.16
1052B-EDU Digital Oscilloscope
50 MHz 1 GS/s
Tek
Trig'd
M Pos: 0.000s
Ch2
Coupling DC
BW Limit On Off 50MHz
Volts/Div Coarse
Probe 10X Voltage
Invert On Off
Maximum 42.0mV
Phase:Ch1 - Ch2 -82.7°?
Frequency 53.53kHz
Phase:Ch2 - Math ?
Maximum 1.44V
Amplitude ?
500mV
50.0mV
M 10.0us
Ch1 ⸝ 10.7mV 53.7099kHz
Oct 28, 2026, 18:24
Menu On/Off
Enabling Teaching
Built-in access to fully customizable courseware content

Figure 4: triangle wave at ≈ 50 kHz